



EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E WEB PROGRAMMING AND INTERFACE

BOOTSTRAP

- A responsive grid system
- Prebuilt components: navbars, cards, buttons
- Utility classes
- JavaScript interactive components

INSTALLING BOOTSTRAP IN REACT

- `npm install bootstrap`

BOOTSTRAP GRID

```
<div className="container">  
  <div className="row">  
    <div className="col-md-4">Column 1</div>  
    <div className="col-md-4">Column 2</div>  
    <div className="col-md-4">Column 3</div>  
  </div>  
</div>
```

BOOTSTRAP CARD EXAMPLE

```
<div className="card">  
    
  <div className="card-body">  
    <h5 className="card-title">Card title</h5>  
    <p className="card-text">Some text.</p>  
  </div>  
</div>
```

REACT + BOOTSTRAP: ONE PAGE LAYOUT

- Header
- Featured Cards
- Footer

BACKEND OVERVIEW (HAPIJS + MONGOOSE)

- `server.js` → server + routes
- `models/Product.js` → schema
- `routes/productRoutes.js` → API endpoints
- `config/db.js` → database connection

MONGODB SETUP

- `npm install @hapi/hapi mongoose --save`

MONGOOSE MODEL (PRODUCT.JS)

```
const mongoose = require("mongoose");
```

```
const productSchema = new mongoose.Schema({  
  name: String,  
  price: Number,  
  category: String  
});
```

```
module.exports = mongoose.model("Product", productSchema);
```

HAPI ROUTES (PRODUCTROUTES.JS)

```
const Product = require("../models/Product");
```

```
module.exports = [  
  {  
    method: "GET",  
    path: "/products",  
    handler: async () => {  
      return await Product.find();  
    }  
  },  
  10  
],
```

HAPI ROUTES (PRODUCTROUTES.JS)

```
method: "POST",  
path: "/products",  
handler: async (req) => {  
  const p = new Product(req.payload);  
  return await p.save();  
}  
}
```

```
];
```

HAPI SERVER (SERVER.JS)

```
const Hapi = require("@hapi/hapi");  
const routes = require("./routes/productRoutes");  
const connectDB = require("./config/db");
```

```
const init = async () => {  
  await connectDB();
```

HAPI SERVER (SERVER.JS)

```
const server = Hapi.server({  
  port: 4000,  
  host: "localhost",  
});  
server.route(routes);  
await server.start();  
console.log("Server running:", server.info.uri);  
};  
init();
```

DATABASE CONNECTION (DB.JS)

```
const mongoose = require("mongoose");

module.exports = async () => {
  await
  mongoose.connect("mongodb://localhost:27017/hapi_demo");
  console.log("MongoDB connected");
};
```

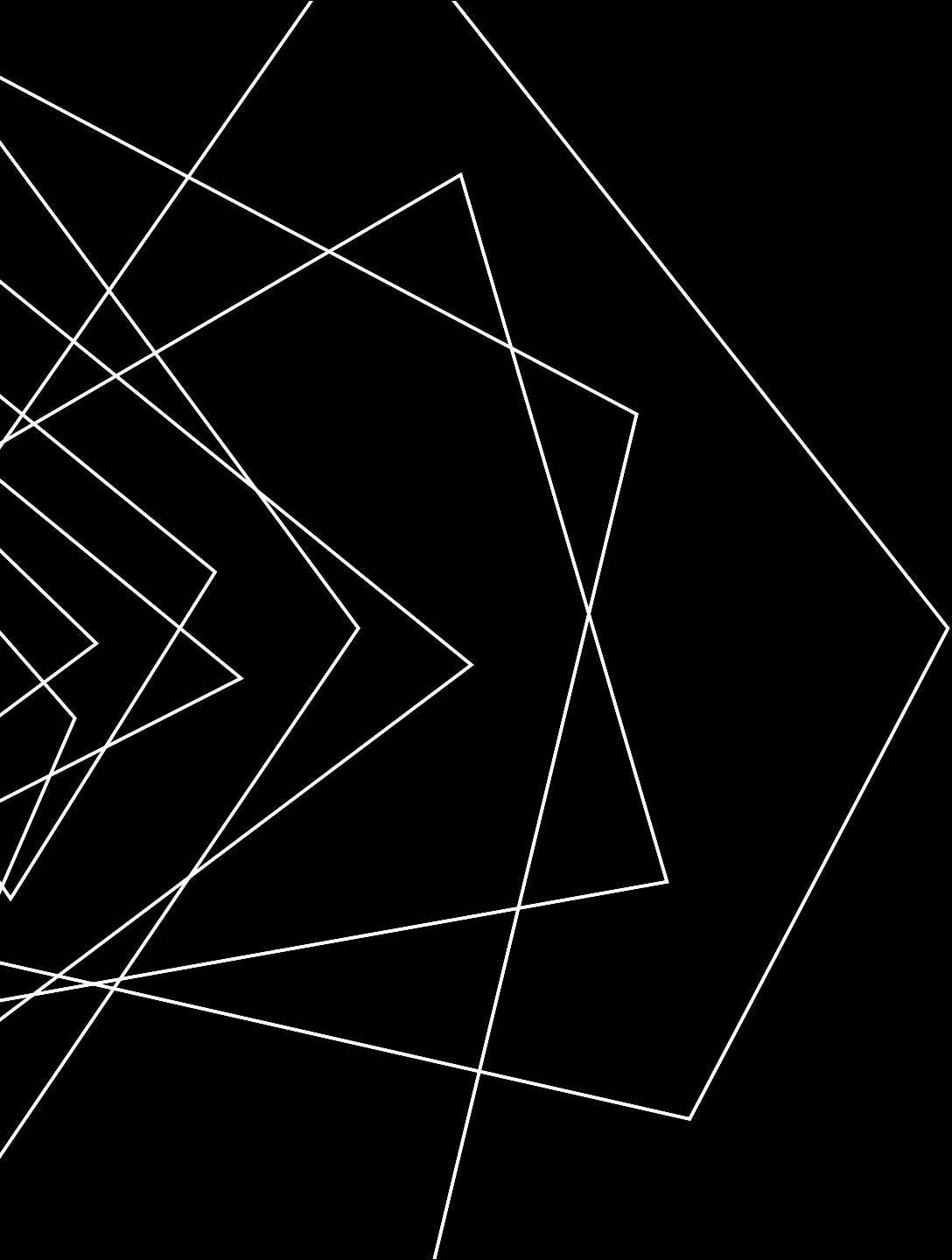
TESTING WITH CURL

- GET

```
curl http://localhost:4000/products
```

- POST

```
curl -X POST http://localhost:4000/products \  
-H "Content-Type: application/json" \  
-d '{"name":"Laptop","price":1200,"category":"Electronics"}'
```



THANK YOU

Yau Ka Cheung
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